

URD ATLAS

Methodology & Data Reference Whitepaper

Blockchain Regime Classification: Architecture, Evidence, Verification, and Traceability

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This document constitutes the canonical public methodology reference for the Urd Atlas data product. It is intended for use in technical due diligence, regulatory review, audit, academic citation, and compliance workflows. It does not contain price data, forecasts, trading signals, or investment recommendations.

Version note: Document version and methodology version are tracked separately. Document version 1.0 is the first whitepaper publication. Methodology version v1.2 is the current methodology contract governing the interpretation of all published artifacts. See Section 18 for changelog.

Abstract

Urd Atlas publishes daily blockchain regime reference data for Bitcoin, Ethereum, Arbitrum, and Base. Each publication cycle produces three artifact layers per chain: Gold (daily on-chain observations), Derived (deterministic rolling transforms of Gold), and Meta (analytical classification layer). The Meta layer exposes a categorical regime label drawn from a five-state vocabulary — STABLE, HEATING, CONGESTED, CHEAP, and UNKNOWN/DEGRADED — together with a three-axis scorecard, a ranked driver set, a confidence score, freshness fields, and a deterministic integrity hash anchoring each named-state publication to its public payload. UNKNOWN/DEGRADED rows do not carry a determinism hash; their identity is anchored by chain, date, methodology version, updated_through, confidence score, and status label.

The product is descriptive and chain-relative. It does not publish price data, asset forecasts, trading signals, or investment recommendations. It is intended for integration into external research pipelines where users require a stable, auditable, machine-readable state variable rather than a dashboard narrative.

Keywords: blockchain analytics, regime classification, on-chain data, confidence gating, deterministic publication, reference data, audit trail, Bitcoin, Ethereum, Arbitrum, Base

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A Appendix A: Sample Artifact Payloads

B Appendix B: Glossary of Public Terms

Reading path

Technical buyers: prioritise Sections 3, 7, 9, 11, and 12.

Compliance and audit reviewers: prioritise Sections 11–15.

Research users: prioritise Sections 8–9 and Appendix A.

1. Introduction and Product Mandate

Urd Atlas is a deterministic, daily-published blockchain regime reference data service covering four networks: Bitcoin (BTC), Ethereum (ETH), Arbitrum (ARB), and Base (BASE). The product addresses one narrow analytical question: does the current on-chain activity pattern still fall within the normal noise range for this chain, or has it shifted into a materially different regime under the published methodology?

The output is reference data, not advisory output. Urd Atlas publishes descriptive, chain-relative classifications. The data product does not publish price observations, return forecasts, trading signals, portfolio recommendations, or any field intended to constitute financial advice. Its outputs are designed as inputs into customers' own downstream analytical processes.

The primary customer is not a casual dashboard reader. The intended user is a research desk, quantitative analyst, data engineer, or advanced independent analyst who wants a daily, joinable state variable for chain conditions and wants that state variable to be versioned, auditable, and sufficiently explicit to support diligence.

Mandate statement: Urd Atlas classifies observed blockchain network state. It does not infer future asset performance. It does not rank assets. It does not tell a user what to buy, sell, hold, or avoid.

2. Design Commitments

The methodology is governed by six design commitments that define the boundary between a reference data product and a discretionary interpretation layer. A data buyer should be able to ask why a label exists, what evidence supported it, whether the label passed the confidence gate, and which methodology version governed the row.

Commitment	Operational meaning
Strict descriptiveness	Labels describe observed chain conditions relative to the chain's own recent history. No cross-chain absolute ranking is implied.
Epistemic conservatism	When evidence is insufficient to support a named classification, the system publishes UNKNOWN/DEGRADED rather than forcing a confident-looking label.
Deterministic publication	Given the same public regime payload, the same named-state hash is produced. Named outputs are anchored to a canonical payload.
Methodology transparency	Thresholds, classification rules, confidence gates, field semantics, and interpretation boundaries are documented publicly.
Versioned contract	Field meaning and analytical semantics are governed by methodology_version. Breaking semantic changes require a version increment.
Archived-as-published principle	Historical artifacts are retained as-published unless an explicit correction is issued and documented in the changelog.

3. Artifact Architecture

Each publication cycle produces three artifact files per chain per UTC calendar day. The layers form a deliberate separation of concerns: observation, deterministic transform, and analytical classification.

Layer	Role	Interpretation rule
Gold	Daily observation layer	Direct daily chain aggregates or robust daily summaries. No regime logic, confidence degradation, or categorical interpretation.
Derived	Deterministic transform layer	Rolling transforms built from Gold, primarily MA7 and MA30 features used for trend context.
Meta	Analytical classification layer	Publishes regime label, confidence, scorecard, driver set, freshness fields, methodology version, and determinism hash (named-state rows only).



Figure 1. Urd Atlas publication pipeline. Each stage is deterministic and separately interpretable.

Separation rationale

Gold answers: what was observed? Derived answers: how does the observation compare to recent temporal context? Meta answers: which regime label is supported by the current evidence surface under the published methodology?

The separation ensures that raw observations, trend transforms, and analytical classifications can each be inspected, cited, and verified independently.

4. Gold Layer

Gold publishes direct daily chain observations or robust daily summaries for each chain. Gold does not apply regime logic, confidence gating, smoothing, or categorical interpretation. It is the measurement layer used by both the published Derived artifact and the Meta classification layer.

Field family	Public meaning	Verification
Daily counts	Daily transaction volume and block production activity, including <code>tx_count_daily</code> and <code>block_count_daily</code> .	Class B
Native value and fee fields	Native-denominated transfer throughput and typical same-day transaction magnitude or fee burden, including <code>median_tx_value_native</code> and <code>median_tx_fee_native</code> .	Class B
Execution-quality and capacity fields	Daily failure burden or capacity usage where semantically meaningful, including <code>failed_tx_rate</code> and <code>gas_utilization_pct</code> .	Class B
Breadth and cadence fields	Participation breadth and typical inter-block interval, including <code>unique_active_addresses</code> and <code>avg_block_time_sec</code> .	Class B

Global interpretation rules

All dates are UTC calendar dates. All analytical interpretation is chain-relative, not cross-chain absolute. No price conversion is applied inside published artifacts. Unsupported or unreliable fields are published as null. If field meaning changes materially, methodology_version must change.

4.1 Example Gold payload

```
{
  "chain": "ethereum",
  "date": "2026-03-31",
  "updated_through": "2026-03-31",
  "tx_count_daily": 1423041,
  "median_tx_fee_native": 0.00342,
  "gas_utilization_pct": 0.962,
  "failed_tx_rate": 0.071,
  "avg_block_time_sec": 12.1,
  "block_count_daily": 7143
}
```

5. Derived Layer

Derived fields are deterministic transforms of Gold. The core public pattern is the rolling average family: `<metric>__ma7` and `<metric>__ma30`. These fields provide context for charting, trend interpretation, and downstream user workflows without altering the public meaning of the original Gold observations.

Transform	Definition	Interpretation
__ma7	7-day simple moving average of the corresponding Gold metric.	Short-term smoothed condition.
__ma30	30-day simple moving average of the corresponding Gold metric.	Medium-term smoothed condition.
Archive-start handling	At the beginning of the archive, available observations are used rather than forcing nulls solely due to insufficient lookback depth.	Preserves continuity at archive start.

Derived is published as a separate artifact layer rather than embedded in Gold because smoothed series are derivative quantities. Separating them makes the derivation explicit and independently auditable, and prevents raw daily observations from being confused with trend-smoothed context values.

5.1 Example Derived payload

```
{
  "chain": "ethereum",
  "date": "2026-03-31",
  "updated_through": "2026-03-31",
  "tx_count_daily__ma7": 1382201.0,
  "tx_count_daily__ma30": 1284004.0,
  "median_tx_fee_native__ma7": 0.00311,
  "median_tx_fee_native__ma30": 0.00227,
  "gas_utilization_pct__ma7": 0.948,
  "gas_utilization_pct__ma30": 0.901
}
```

}

6. Meta Layer

Meta is the primary commercial deliverable. It translates the observation and transform layers into a named state through explicit evidence fields rather than opaque natural-language interpretation. A Meta row is designed to be joined into a user's downstream dataset and filtered by confidence, chain, ruleset, freshness, and methodology version.

Component	Representative fields	Purpose
Regime state	status.label, regime.label, status.one_liner	Top-line state and compact explanation.
Confidence	confidence.confidence_score, data_quality_score, label_confidence_score	Evidence strength and publication eligibility.
Publication gate	publish_confidence.threshold	Named-state gate value (0.40).
Scorecard	scorecard.dimensions	Demand / Friction / Capacity display scores.
Drivers	regime.drivers[]	Ranked evidence variables with z-score, percentile, momentum, current value.
Freshness	updated_through, lag_days_vs_utc_today	Timeliness of supporting evidence.
Methodology	methodology_version, ruleset_id, window_days	Versioned interpretation contract.
Integrity	regime.determinism_hash	Named-state payload anchor. Present only on named-state rows. Not present on UNKNOWN/DEGRADED rows.

Meta is the only interpretive layer. The interpretation is constrained by rules, thresholds, and confidence gates. It is not a discretionary commentary channel.

7. Confidence System

The confidence system is the primary epistemic control in the Urd Atlas publication pipeline. It serves two distinct functions: it gates publication eligibility, preventing labels from being published when the evidence surface is too weak; and it degrades displayed scorecard scores toward neutral when confidence is reduced but remains above the gate threshold.

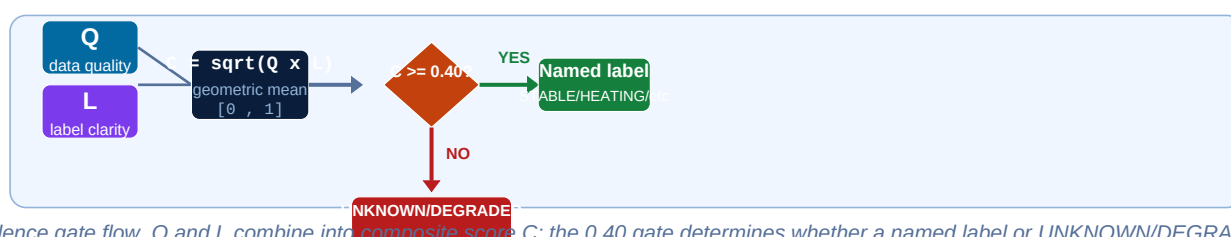


Figure 2. Confidence gate flow. Q and L combine into composite score C; the 0.40 gate determines whether a named label or UNKNOWN/DEGRADED is published.

7.1 Composite confidence score

The current published confidence score is the geometric mean of two component scores:

Component	Symbol	Definition
Data quality score	Q	Measures data sufficiency: completeness of Gold metrics, freshness relative to the observation date, density of recent history, and coverage of required metric set. Range: [0, 1].
Label confidence score	L	Measures analytical clarity: how strongly the current evidence surface distinguishes the published label from the nearest alternative classification. Range: [0, 1].
Composite confidence score	C	$C = \sqrt{Q \times L}$. Both dimensions must be adequate for the composite to be high. The geometric mean is intentionally conservative: strong label clarity cannot fully offset poor data quality, and vice versa. Range: [0, 1].

7.2 Publication gate and score degradation

The canonical publication gate threshold is **0.40**. When the composite confidence score falls below this threshold, the product publishes UNKNOWN/DEGRADED as the regime label regardless of what the underlying axis scores indicate.

Gate semantics

The confidence gate is not a statement that raw data is wrong. It is a publication eligibility rule: the evidence surface is not strong enough to support a named descriptive classification at this time. Raw Gold fields remain available even when Meta publishes UNKNOWN/DEGRADED.

Many analytics systems either suppress uncertainty or present a continuous label surface. Urd Atlas instead exposes insufficient evidence as an explicit publication state, making the absence of confidence visible and machine-readable.

Band	Score range	Interpretation
Good	≥ 0.70	Label is well-supported by the current evidence surface. Read normally.
Caution	0.40 – 0.69	Label is still published, but scorecard scores are pulled toward neutral in proportion to the confidence deficit. Cross-check before drawing strong conclusions.
Degraded	< 0.40	Publication gate blocks a named label. UNKNOWN/DEGRADED is published regardless of axis structure. No determinism hash is issued for these rows.

Scorecard confidence degradation formula

$$\text{published_score} = 50 + (\text{score_raw} - 50) \times \text{effective_confidence}$$

At confidence = 1.0, the published score equals the raw score. At confidence = 0.0, the published score equals 50 (fully neutral). Intermediate values produce proportionally attenuated scores. Customers can directly verify the published display score from the two published row inputs score_raw and effective_confidence (Class A verifiable).

8. Scorecard Architecture

The scorecard compresses current chain conditions into three analytical axes: Demand, Friction, and Capacity. Scores are chain-relative and bounded to a 0–100 display scale with 50 as the neutral point representing typical recent conditions for that specific chain. The scorecard is not a separate label engine; it is an evidence surface that the regime classification engine reads alongside driver evidence.

8.1 Axes: Demand, Friction, Capacity

Axis	Primary signal	Key metrics
Demand	Transaction throughput and network utilization pressure	tx_count_daily, unique_active_addresses
Friction	Cost burden relative to value transferred	median_tx_fee_native; fee_burden_proxy = median_tx_fee_native / median_tx_value_native
Capacity	Network capacity utilization and block production regularity	gas_utilization_pct (EVM chains); blocktime_instability proxy (Bitcoin only)

8.2 Normalization and score construction

Score construction uses robust normalization against each chain's own historical baseline. The currently implemented score family applies 7-day smoothing before historical comparison and excludes the most recent 14 days from the baseline window.

Step	Operation	Formula / parameter
1 — Input smoothing	7-day rolling average applied to the raw Gold metric series	MA7(metric), window = 7 days
2 — Baseline	365-day baseline, excluding the most recent 14 days	Window: 365d, holdout: 14d
3 — Robust z-score	MAD-based normalization of the smoothed series against baseline	$z = (x - \text{median}) / (1.4826 \times \text{MAD})$
4 — Bounded display score	Monotone mapping of z into [0, 100] via tanh compression	$\text{score_raw} = 50 + 40 \times \tanh(z / 1.5)$
5 — Confidence degradation	Pull published score toward neutral by confidence	$\text{published} = 50 + (\text{score_raw} - 50) \times \text{eff_conf}$

Important: scorecard z-score and driver z-score are distinct calculations

regime.drivers[].z_robust is computed from 180-day raw daily values using $0.6745 \times (x - \text{median}) / \text{MAD}$. Scorecard dimension scores are computed from 7-day rolling averages against a 365-day baseline using $(x - \text{median}) / (1.4826 \times \text{MAD})$. These are separate calculations with separate input series, windows, and analytical purposes. They will not produce identical values for the same metric on the same day.

8.3 Bitcoin capacity proxy

For Bitcoin, gas_utilization_pct is not published because Bitcoin does not expose EVM-style gas semantics. The Capacity axis therefore relies entirely on blocktime_instability:

Bitcoin capacity proxy formula

$$\text{blocktime_instability} = \frac{\text{rolling_mean}(|\text{avg_block_time_sec} - \text{rolling_median_30}(\text{avg_block_time_sec})|}{\text{rolling_median_30}(\text{avg_block_time_sec})}$$

This index measures deviation from the chain's own recent block-time norm in both directions. It is a volatility signal, not a directional slow-block measure.

Bitcoin capacity interpretation boundary

A period of consistently fast block production and a period of consistently slow block production can both produce low instability scores if both are stable relative to the recent norm. A high BTC Capacity score means block timing is unusually erratic, not that blocks are necessarily slow. Read BTC Capacity as a stress-or-instability proxy, not as a directional congestion indicator.

9. Regime Classification Engine

The classification engine maps the current evidence surface to a mutually exclusive label. The vocabulary is intentionally small because the product is designed to publish stable reference states rather than a dense taxonomy of narrative conditions. A CONGESTED label on Ethereum and a CONGESTED label on Arbitrum do not imply identical absolute conditions: each chain meets the published criteria relative to its own calibration and ruleset.

9.1 Five-state label vocabulary

Label	Plain meaning	Trigger condition (summary)
STABLE	No axis is in a notably elevated or depressed state.	None of CONGESTED, CHEAP, or HEATING conditions hold.
HEATING	Demand is elevated and directional trend confirms acceleration.	Demand axis HIGH and at least one axis shows positive momentum ($\text{momentum_7d_vs_30d} \geq 0.15$).
CONGESTED	Network is under significant pressure.	Capacity EXTREME_HIGH, or (Capacity HIGH and Friction HIGH).
CHEAP	Both friction and capacity pressure are low.	Friction LOW and Capacity LOW.
UNKNOWN/DEGRADED	Evidence quality is insufficient for a named label.	Confidence score < 0.40, evaluated before all other rules.

9.2 Threshold band assignment

The classification engine assigns each axis to a band using two parallel criteria: a 90-day percentile rank and a robust z-score. The band fires when **either** criterion is met (OR logic). This allows detection of both distributional extremes and robust statistical deviations.

Band	Percentile	Z-score	Logic	Role in regime
EXTREME_HIGH	$\geq 95\text{th}$	≥ 2.5	OR	Triggers CONGESTED alone (Capacity axis only)

HIGH	≥ 80th	≥ 1.5	OR	Feeds CONGESTED (Capacity+Friction) and HEATING (Demand)
NORMAL	20–80th	–1.5 to +1.5	—	Default — no band condition fires
LOW	≤ 20th	≤ –1.5	OR	Feeds CHEAP (Friction+Capacity both LOW)
EXTREME_LOW	≤ 5th	≤ –2.5	OR	Stronger low signal — also feeds CHEAP

Percentile rank is computed against the trailing 90-day window. Driver robust z-score is computed from 180-day raw daily history using $0.6745 \times (x - \text{median}) / \text{MAD}$. These windows intentionally differ from scorecard display normalization and should not be treated as interchangeable.

9.3 Label evaluation order

Labels are evaluated in a fixed priority order. Once a condition is satisfied, subsequent checks are skipped. This priority model ensures mutual exclusivity and deterministic behavior.



Figure 3. Regime evaluation flow. The confidence gate always runs first. The first satisfied condition wins.

9.4 Label stability and persistence properties

Urd Atlas does not apply a universal fixed multi-day confirmation window across all regime labels. Persistence properties differ by label:

- **HEATING** requires a directional trend condition: $\text{momentum_7d_vs_30d} \geq 0.15$. HEATING will not appear on a single-day elevation alone.
- **CONGESTED** and **CHEAP** do not require trend confirmation. Either label can fire on a single-day threshold crossing.
- **STABLE** fires as the residual default.

Downstream consumer warning: label volatility

Labels can change day to day in response to daily evidence. Consumers using regime labels as period classifiers for backtesting or segmentation should apply their own minimum-duration filter if multi-day stability is required. Urd Atlas does not apply such a filter in the published output.

10. Freshness Policy

Freshness and confidence are related but orthogonal dimensions. Freshness describes how recently a row has been published relative to expected publication cadence. Confidence describes how much evidence supports the analytical state of that row. A row can be on schedule and still low-confidence. A row can be delayed and still mathematically valid as the latest available state.

Chain	Expected lag	Soft warning	Hard fail
Bitcoin	~1 day	> 2 days	> 4 days
Ethereum	~1 day	> 2 days	> 4 days
Arbitrum	~7 days	> 10 days	> 15 days
Base	~7 days	> 10 days	> 15 days

The 7-day cadence for Arbitrum and Base is an intentional design condition reflecting data availability and pipeline architecture. Observing a 7-day lag for ARB or BASE is normal; it is not a pipeline failure.

Freshness field	Meaning
updated_through	Latest Gold observation date actually available to the Meta calculation.
lag_days_vs_asof_date	Publication-time freshness relative to the row date.
lag_days_vs_utc_today	Runtime freshness relative to the current UTC date. This field is computed at runtime and will differ between the time of publication and the time of reading.

11. Release Integrity and Determinism

11.1 Determinism hash construction

Every **named-state** Meta row — that is, every row where the published label is STABLE, HEATING, CONGESTED, or CHEAP — carries a `regime.determinism_hash`. UNKNOWN/DEGRADED rows do **not** carry a determinism hash. Their public identity is anchored by the fields listed in Section 11.2.

Determinism hash construction specification

```
payload = {
  'asof_date': meta['regime']['asof_date'],
  'chain': meta['chain'],
  'drivers': meta['regime']['drivers'],
  'label': meta['regime']['label'],
  'ruleset_id': meta['regime']['ruleset_id']
}
canonical = json.dumps(payload, sort_keys=True, separators=(',', ':'))
hash = hashlib.sha256(canonical.encode('utf-8')).hexdigest()[:12]
```

Given the same payload, the same 12-character string is always produced. If the named regime payload changes, the hash changes. If it does not change, the hash remains stable. The hash anchors the public named-state payload — it is not a claim that all upstream internals are disclosed.

11.2 Public row identity model

Public row identity is anchored in fields present in the published archive, not in a separate hidden revision integer.

Row type	Public identity fields
All Meta rows	chain, date, methodology_version
Named-state rows (STABLE, HEATING, CONGESTED, CHEAP)	chain, date, methodology_version, regime.determinism_hash
UNKNOWN/DEGRADED rows	chain, date, methodology_version, updated_through, confidence.confidence_score, status.label

UNKNOWN/DEGRADED identity clarification

UNKNOWN/DEGRADED rows do not carry a determinism hash because the regime payload (label, drivers, ruleset_id) is not meaningful in a degraded state — drivers are empty and the label is determined solely by the confidence gate, not by axis structure. The public identity of these rows is instead anchored by the fields listed above, all of which are present in the published artifact and verifiable by the reader.

11.3 Archived-as-published principle

Historical outputs are intended to be interpreted as valid under the methodology version under which they were published. If methodology changes in a way that changes field meaning or label semantics, that change is versioned rather than silently applied retroactively. The changelog (Section 18) records all customer-facing changes classified by type, backward-compatibility impact, methodology version bump, archival impact, and subscriber action.

12. Verification and Evidence Paths

12.1 Verification classification

Class	Definition	Typical examples
A — Directly reproducible	Can be reproduced from published artifacts alone, without access to any external data source.	determinism_hash, scorecard display score, Gold→Meta driver consistency, confidence gate, published label identity
B — Independently checkable	Can be verified against public chain evidence but cannot be fully reconstructed from Urd Atlas artifacts alone.	tx_count_daily, block_count_daily, fee direction, block-time behaviour
C — Black box by design	Publicly interpretable but not fully reconstructable. Intentionally withheld to prevent source-data reconstruction or pipeline cloning.	Private calibration constants, intermediate feature tables, exact upstream join logic

Diligence boundary

Urd Atlas is designed to make published outputs auditable without pretending the entire production classifier is open-sourced. Reproducibility of the public trust layer and full disclosure of every proprietary internal are different concerns. The classification above makes this explicit.

12.2 End-to-end diligence path

The following six-step path constitutes the intended end-to-end diligence flow for a technical buyer using the public sample pack (ethereum / 2026-03-31).

Step	File or action	Expected outcome	Class
1	ethereum/2026-03-31/gold.json	Gold row loads with chain = ethereum, date = 2026-03-31.	A / B
2	ethereum/2026-03-31/derived.json	Derived row loads for same chain and date, exposing __ma7 and __ma30.	A
3	ethereum/2026-03-31/meta.json	Meta row resolves to CONGESTED with gate threshold 0.40.	A
4	Recompute determinism hash	Hash recomputes to 81b295000696.	A
5	Independently inspect Gold-level facts	Counts and fee behaviour checkable against public Ethereum chain evidence.	B
6	Interpret remaining internals	Full private classifier internals remain deliberately non-public.	C

12.3 Executable verification script

The following Python script can be executed against the public sample pack to verify the complete diligence path. All assertions should pass on the unmodified ethereum/2026-03-31 sample files.

```
import hashlib, json
from pathlib import Path

base = Path("sample-pack/ethereum/2026-03-31")
gold = json.loads((base / "gold.json").read_text(encoding="utf-8"))
derived = json.loads((base / "derived.json").read_text(encoding="utf-8"))
meta = json.loads((base / "meta.json").read_text(encoding="utf-8"))

# 1. Basic artifact identity
assert gold["chain"] == "ethereum"
assert gold["date"] == "2026-03-31"
assert derived["chain"] == "ethereum"
assert derived["date"] == "2026-03-31"
assert meta["chain"] == "ethereum"
assert meta["date"] == "2026-03-31"

# 2. Gold -> Meta current-value consistency for named drivers
for driver in meta["regime"]["drivers"]:
    metric = driver["metric"]
    if metric in gold and "current" in driver:
        assert driver["current"] == gold[metric]

# 3. Required Derived fields present
for field in ["tx_count_daily__ma7", "tx_count_daily__ma30",
              "median_tx_fee_native__ma7", "median_tx_fee_native__ma30"]:
    assert field in derived, field

# 4. Publication gate and named output
assert meta["status"]["label"] == "CONGESTED"
```

```
assert meta["regime"]["label"] == "CONGESTED"
assert meta["publish_confidence"]["threshold"] == 0.40
assert meta["regime"]["ruleset_id"] == "eth_l1_v1"

# 5. Recompute determinism_hash
payload = {
    "asof_date": meta["regime"]["asof_date"],
    "chain": meta["chain"],
    "drivers": meta["regime"]["drivers"],
    "label": meta["regime"]["label"],
    "ruleset_id": meta["regime"]["ruleset_id"],
}
canonical = json.dumps(payload, sort_keys=True, separators=(",", ":"))
sha = hashlib.sha256(canonical.encode("utf-8")).hexdigest()[:12]
assert sha == meta["regime"]["determinism_hash"]
assert sha == "81b295000696"

print("All assertions passed.")
print({"chain": meta["chain"],
      "date": meta["date"],
      "label": meta["status"]["label"],
      "gate_threshold": meta["publish_confidence"]["threshold"],
      "determinism_hash": sha})
```

Expected output field	Expected value
chain	ethereum
date	2026-03-31
label	CONGESTED
gate_threshold	0.40
determinism_hash	81b295000696

13. Disclosure Boundaries

The public methodology is designed to make the product auditable in meaning without enabling source-data reconstruction or pipeline cloning. A clear disclosure boundary makes the non-public portion of the product disciplined rather than suspicious.

Public methodology discloses	Public methodology intentionally does not disclose
Field meaning, artifact ownership, and data contract.	Exact upstream AWS schemas or source join logic.
Interpretation of confidence, scorecard, regime, and freshness.	Intermediate feature tables and ingestion repair rules.
All publicly relevant thresholds and gate values.	Enough detail to reconstruct raw source rows from published aggregates.
Worked verification examples for deterministic and gated outputs.	Enough implementation detail to clone the private pipeline.

Classification rules, label evaluation order, and band thresholds.	Full proprietary calibration constants required to reproduce the classifier end-to-end.
Freshness policy and chain-specific publication cadence.	
Provenance model and changelog classification scheme.	

14. Analytical Interpretation Boundaries

The following boundaries define what Urd Atlas outputs should and should not be used to infer. The analytical boundary is part of the product: the system is useful because it is explicit about what it can support and what it declines to support.

Boundary	Operational consequence
Chain-relative only	All labels and scores are chain-relative. A CONGESTED label on Arbitrum does not imply the same absolute state as CONGESTED on Ethereum. Cross-chain comparisons must account for chain-specific calibration (<code>ruleset_id</code>).
Labels can change day to day	CONGESTED and CHEAP can fire from single-day threshold crossings. Consumers requiring multi-day stability must apply their own filters.
Scorecard z-scores and driver z-scores differ	They use different input series, windows, and normalization constants. Do not expect them to match.
Bitcoin Capacity is an instability proxy	It measures deviation from recent block-time norm in either direction, not directional slow-block congestion.
Friction can be elevated without high absolute fees	<code>fee_burden_proxy</code> is a ratio. Friction may increase because transferred value is low, not because nominal fees are high.
No price, forecast, or recommendation	No field constitutes a price observation, return forecast, trading signal, or investment recommendation. All outputs are descriptive and chain-relative.

15. Provenance and Revision Model

Provenance concept	Public meaning	When it changes
<code>date</code>	UTC day the published row describes.	Only if row identity changes.
<code>updated_through</code>	Most recent Gold observation available to the Meta calculation.	When the effective freshest supporting row changes.
<code>methodology_version</code>	Public analytical methodology version governing interpretation.	When field meaning or analytical semantics change materially.
<code>regime.determinism_hash</code>	Named-row integrity anchor. Present only on named-state rows.	When the named regime payload changes.
Dataset / published revision	Publication batch identity at the dataset layer.	When a new publication batch is issued.

Change class	Methodology bump?	Historical rows changed?	Subscriber action?
Docs-only clarification	No	No	No
Pipeline rebuild without methodology change	No	Only if corrected artifacts are republished	No unless correction notice says otherwise
Methodology-breaking analytical change	Yes	Only if explicitly recomputed and republished	Yes — treat pre- and post-bump outputs as version-distinct
Retroactive correction / archival replacement	Case-specific	Yes, for affected rows only	No for most use; correction note should be read

16. AI Use and Quality Controls

Parts of the software development workflow for Urd Atlas may be AI-assisted. Trust in the published product is therefore anchored not in authorship claims, but in controls: versioned public methodology, archived published artifacts, deterministic publication rules, release checks, and public verification paths.

- Versioned public methodology governing what outputs mean
- Archived published artifacts retained verbatim for inspection
- Deterministic publication rules with named-row hash anchors
- Release controls requiring deterministic test cases before publication
- Public verification paths (Section 12) enabling independent reproduction

Release-control rule

Where analytical logic affects published outputs, that logic is validated against deterministic test cases, known input-output expectations, and release checks before publication. This rule applies regardless of whether code was written by a human, AI-assisted, or both.

17. Field Reference Summary

17.1 Key Gold fields

Field	Type	Meaning	V.
tx_count_daily	integer	Confirmed daily transaction count.	B
block_count_daily	integer	Blocks produced on the UTC day.	B
median_tx_fee_native	float	Typical same-day fee in native denomination (median, not mean).	B
median_tx_value_native	float	Typical same-day transaction value. Used in friction proxy.	B
avg_block_time_sec	float	Typical daily inter-block interval. Robust typical interval, not strict mean.	B

gas_utilization_pct	float null	Gas utilization as fraction of block gas limit. Null for Bitcoin.	B
failed_tx_rate	float null	Fraction of transactions that failed. Null where not semantically applicable.	B
unique_active_addresses	int null	Participation breadth. Null where not reliably available.	B

17.2 Key Meta fields

Field	Type	Meaning	V.
status.label	string	Top-line regime state: STABLE, HEATING, CONGESTED, CHEAP, UNKNOWN/DEGRADED.	A
confidence.confidence_score	float	Composite confidence [0,1]. Gate threshold: 0.40.	A
publish_confidence.threshold	float	Canonical gate threshold (0.40). Verifiable from artifact.	A
regime.determinism_hash	string absent	12-char SHA-256 prefix over public regime payload. Present on named-state rows only. Absent on UNKNOWN/DEGRADED rows.	A
regime.ruleset_id	string	Pins output to a specific public ruleset version (e.g. eth_l1_v1).	A
regime.drivers[]	array	Ranked driver set: axis, metric, z_robust (180d raw), pct_90d, momentum_7d_vs_30d, current value. Empty array on UNKNOWN/DEGRADED rows.	A/C
scorecard.dimensions[].score	float	Confidence-degraded display score [0,100]. Verifiable: $50 + (\text{score_raw} - 50) \times \text{eff_conf}$.	A
methodology_version	string	Methodology version governing interpretation.	A
updated_through	date	Latest Gold observation available to this Meta calculation.	A

18. Changelog and Versioning Policy

The following table records all public customer-facing changes to methodology, interpretation, thresholds, contracts, and historical corrections. Docs-only changes are clearly distinguished from output-affecting corrections or methodology-breaking changes.

Date	Class	Affected	Meth.?	Rows?	Action?	Summary
2026-04-21	docs-only	Methodology hub	No	No	No	Introduced structured methodology hub and cross-linked trust-layer pages.

2026-04-21	interpretation-only	Reference / fields / boundaries	No	No	No	Clarified label volatility, regime-vs-scorecard normalization, fee_burden_proxy semantics, BTC capacity as instability proxy.
2026-04-21	contract clarification	Schema / provenance docs	No	No	No	Removed public reliance on a required separate revision integer; aligned provenance with fields present in archive.
2026-04-21	operational docs	Service / API docs / pricing	No	No	No	Added service expectations, public sample pack, and common workflows for pre-purchase diligence.
2026-05-03	methodology consolidation	Whitepaper / public methodology	Yes → v1.2	No	Yes — cite v1.2 going forward	Consolidated public methodology, confidence, scorecard, regime rules, verification, provenance, and sample payload documentation into canonical whitepaper. Clarified that UNKNOWN/DEGRADED rows do not carry a determinism_hash. Added three methodology figures. Toned down language around competing analytics systems.

19. Conclusion

Urd Atlas is a deterministic regime reference data layer for on-chain market state analysis. Its value is not that it replaces researcher judgment, but that it gives researchers a documented, versioned, auditable state variable that can be joined into their own pipelines.

The system is intentionally narrow. It does not publish price data, does not forecast, and does not recommend actions. It publishes descriptive classifications of chain-relative conditions and exposes enough evidence, confidence, freshness, and provenance metadata to support technical due diligence.

For a research-grade user, the critical product property is not narrative persuasion. It is repeatability. Urd Atlas is designed so that a customer can ingest a row, inspect the evidence surface, check the confidence gate, verify the public hash, and preserve the row under the methodology version that governed its interpretation.

Appendix A: Sample Artifact Payloads

The following payloads are representative examples from the public sample pack. Full files are available for download at urdatlas.com/api-docs/samples.

Note on freshness fields in sample payloads: sample payloads are archived-as-generated fixtures, not live runtime snapshots. The field `lag_days_vs_utc_today` shown in sample payloads reflects the value at the time the fixture was generated. When the same field is read from a live API response, it will reflect the current UTC date at the time of the request. Do not read sample freshness values as current runtime freshness.

A.1 Gold layer — Ethereum, 2026-03-31

```
{
  "chain": "ethereum",
  "date": "2026-03-31",
  "updated_through": "2026-03-31",
  "tx_count_daily": 1423041,
  "median_tx_fee_native": 0.00342,
  "gas_utilization_pct": 0.962,
  "failed_tx_rate": 0.071,
  "avg_block_time_sec": 12.1,
  "block_count_daily": 7143
}
```

A.2 Derived layer — Ethereum, 2026-03-31

```
{
  "chain": "ethereum",
  "date": "2026-03-31",
  "updated_through": "2026-03-31",
  "tx_count_daily_ma7": 1382201.0,
  "tx_count_daily_ma30": 1284004.0,
  "median_tx_fee_native_ma7": 0.00311,
  "median_tx_fee_native_ma30": 0.00227,
  "gas_utilization_pct_ma7": 0.948,
  "gas_utilization_pct_ma30": 0.901
}
```

A.3 Meta layer — named state, CONGESTED — Ethereum, 2026-03-31

```
{
  "chain": "ethereum",
  "date": "2026-03-31",
  "updated_through": "2026-03-31",
  "methodology_version": "v1.2",
  "status": {
    "label": "CONGESTED",
    "one_liner": "Demand, fees, and utilization remain elevated versus recent history."
  },
  "publish_confidence": { "threshold": 0.40 },
  "confidence": {
    "confidence_score": 0.847,
    "data_quality_score": 0.91,
    "label_confidence_score": 0.79,
    "lag_days_vs_utc_today": 0
  },
}
```

```

"regime": {
  "label": "CONGESTED",
  "asof_date": "2026-03-31",
  "ruleset_id": "eth_l1_v1",
  "window_days": 180,
  "determinism_hash": "81b295000696",
  "drivers": [
    { "axis": "demand", "metric": "tx_count_daily",
      "z_robust": 2.18, "pct_90d": 0.97,
      "momentum_7d_vs_30d": 0.44, "current": 1423041 },
    { "axis": "friction", "metric": "median_tx_fee_native",
      "z_robust": 2.41, "pct_90d": 0.99,
      "momentum_7d_vs_30d": 0.58, "current": 0.00342 },
    { "axis": "capacity", "metric": "gas_utilization_pct",
      "z_robust": 1.84, "pct_90d": 0.95,
      "momentum_7d_vs_30d": 0.21, "current": 0.962 }
  ]
}

```

A.4 Meta layer — UNKNOWN/DEGRADED — Ethereum, 2025-04-21

Note: UNKNOWN/DEGRADED rows do not carry a `determinism_hash`. Row identity is anchored by chain, date, methodology version, updated through, confidence score, and status.label.

```

{
  "chain": "ethereum",
  "date": "2025-04-21",
  "updated_through": "2025-04-21",
  "methodology_version": "v1.2",
  "status": {
    "label": "UNKNOWN/DEGRADED",
    "one_liner": "Evidence quality is below the named-state publication threshold."
  },
  "publish_confidence": { "threshold": 0.40 },
  "confidence": {
    "confidence_score": 0.318,
    "data_quality_score": 0.41,
    "label_confidence_score": 0.25,
    "lag_days_vs_utc_today": 8
  },
  "regime": {
    "label": "UNKNOWN/DEGRADED",
    "asof_date": "2025-04-21",
    "ruleset_id": "eth_l1_v1",
    "window_days": 180,
    "drivers": []
  }
}

```

A.5 Meta layer — HEATING — Arbitrum, 2026-03-25

```

{
  "chain": "arbitrum",
  "date": "2026-03-25",
  "updated_through": "2026-03-25",

```

```
"methodology_version": "v1.2",
"status": {
  "label": "HEATING",
  "one_liner": "Activity is firming relative to recent chain norms but not yet congested."
},
"publish_confidence": { "threshold": 0.40 },
"confidence": {
  "confidence_score": 0.732,
  "data_quality_score": 0.81,
  "label_confidence_score": 0.66,
  "lag_days_vs_utc_today": 0
},
"regime": {
  "label": "HEATING",
  "asof_date": "2026-03-25",
  "ruleset_id": "arb_v1",
  "window_days": 180,
  "determinism_hash": "da96a075f205",
  "drivers": [
    { "axis": "demand", "metric": "tx_count_daily",
      "z_robust": 1.36, "pct_90d": 0.83,
      "momentum_7d_vs_30d": 0.17, "current": 1182403 }
  ]
}
```

Appendix B: Glossary of Public Terms

Terms are defined as they are used within Urd Atlas. Definitions are product-specific and describe how each term is used in the published methodology, not how every other analytics product necessarily uses the same term.

Term	Definition
Gold	Daily observation layer. Direct chain metrics and robust daily summaries without regime logic, confidence gating, or categorical interpretation.
Derived	Deterministic rolling transforms of Gold metrics, primarily MA7 and MA30.
Meta	Classification layer containing regime label, confidence, scorecard, drivers, freshness, and provenance fields.
Named-state row	A Meta row whose published label is STABLE, HEATING, CONGESTED, or CHEAP. Named-state rows carry a determinism hash. UNKNOWN/DEGRADED rows do not.
Confidence gate	The publication threshold (0.40) that blocks named labels when the composite confidence score is below the threshold.
Determinism hash	A 12-character SHA-256 prefix over the canonical public regime payload. Serves as an integrity anchor for named-state rows only.
Ruleset ID	A public identifier tying a row to a chain-specific classification ruleset (e.g. eth_l1_v1, arb_v1).
Chain-relative	Scores and labels are interpreted relative to the chain's own historical distribution, not as absolute cross-chain rankings.
fee_burden_proxy	$\text{median_tx_fee_native} / \text{median_tx_value_native}$. A dimensionless ratio measuring fee cost relative to transferred value, not absolute fee size.
blocktime_instability	Bitcoin-specific capacity proxy measuring rolling deviation from the chain's own recent block-time norm in either direction.
Verification Class A	Directly reproducible from published artifacts alone.
Verification Class B	Independently checkable against public chain evidence but not fully reconstructable from Urd Atlas artifacts alone.
Verification Class C	Publicly interpretable but intentionally not fully reconstructable, to prevent source-data reconstruction or pipeline cloning.
Archived-as-published	The principle that historical artifacts are retained as originally published. Any corrections are explicitly disclosed in the changelog.

End of document. For the latest version of this whitepaper and for full sample pack access, visit urdatlas.com. Technical questions: support@urdatlas.com (response within 2–4 business days).